

Dr. Arti Dwivedi

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Objective

To pursue an academically enriching career where I can contribute meaningfully to the fields of Physics and interdisciplinary teaching, particularly in Nonlinear Dynamics and Indian Knowledge Systems, while nurturing scientific temperament among students.

Academic Qualifications

Degree	Institution	Year	Percentage
Ph.D. (Physics)	Central University of Allahabad, U.P., India	2015	—
M.Sc. (Physics)	C.S.J.M. University, Kanpur, U.P., India	2007	58%
B.Sc. (PCM)	C.S.J.M. University, Kanpur, U.P., India	2005	58%
Intermediate (PCM)	P.D. Khanna Girls Inter College, Kanpur, U.P.	2002	62%
High School (Science)	G.N.G. Girls Inter College, Kanpur, U.P.	2000	55%

Teaching Experience (9 Years 8 Months)

- Assistant Professor (Physics)**
Thakur College of Engineering & Technology (TCET), Mumbai
(Dec 2022 – Mar 2025)
 - Taught core Physics subjects in undergraduate engineering programs
 - Handled Indian Knowledge System (IKS) curriculum** from **Aug 2023 to Mar 2025** as part of NEP-based interdisciplinary module
 - Delivered lectures on Indian holistic systems thinking
- Guest Faculty**
Central University of Allahabad, Department of Physics
(Aug 2018 – Apr 2020)
- Teaching Assistant (during Ph.D.)**
Central University of Allahabad
(2009 – 2015)

Ph.D. Research

Thesis Title: *Chaos, Fractals and Their Applications in Electronics and Other Disciplines: Chaos-based Secure Communication*

Degree Awarded By: Central University of Allahabad, 2015

Examiners:

- Prof. Ashok K. Mittal – Central University of Allahabad
 - Prof. Sudeshna Sinha – IISER Mohali
 - Dr. Santo Banerjee – INSPEM, University Putra Malaysia
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Fellowships

- UGC-CRET Fellowship (2009–2011)
 - DST Project JRF (2011–2013)
 - CSIR SRF (2013–2015)
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Research Publications (International Journals)

Total Citations: 36 | **h-index:** 3 | **i10-index:** 2

1. **Adaptive Synchronization of Diffusion-less Lorenz Systems and Secure Communication of Digital Signals by Parameter Modulation**, *IET Communications*, 2012. [Scopus]
 2. **An Improved Secure Communication Scheme Using Adaptive Parameter Synchronization**, *Springer CCIS Series*, 2013. [Scopus]
 3. **Parameter Adaptation Technique for Rapid Synchronization and Secure Communication**, *European Physical Journal – Special Topics*, Springer. [Scopus]
 4. **Secure Communication Based on Chaotic Switching and Rapid Synchronization Using Parameter Adaptation**, *Int. J. of Innovative Computing, Information and Control*, 2014. [Scopus]
 5. **Chaotic Communication Using Pecora-Carroll Complete Replacement and Parameter Modulation Without Controller**, *IEEE SCES Conference Proceedings*, 2014. [Scopus]
 6. **Secure Communication Using Four-Wing Hyperchaotic Attractor**, *Springer CCIS Series*, 2014. [Scopus]
 7. **Role of Parameter Adaptation in Chaotic Communication**, *International Journal of Trust Management in Computing and Communications*, 2016. [Web of Science, UGC CARE]
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Conferences & Presentations (Selected)

- IEEE SCES 2014 – *Chaotic Communication using Pecora-Carroll Replacement*
 - ICACCI-SSCC 2013 – *Adaptive Synchronization for Secure Communication*
 - CNSD 2015 – *Secure Communication Using Arbitrary Chaotic Systems*, IISER Mohali
 - MULTICON-W 2023, TCET – *Whispering Stars: A Sneak Peek into Multi-messenger Astronomy*
 - Experimental Chaos and Complexity Conference, USA – *(Abstract submitted, 2012)*
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Workshops & Training Programs

- **SERC School on Nonlinear Dynamics**, Univ. of Delhi (DST Sponsored) – 2009
 - **Scientific Paper Writing**, NASI Allahabad – 2011
 - **Statistical Methods for Engineers**, IIT Kanpur – 2013
 - **Statistical Analysis using R**, Central University of Allahabad – 2013
 - **NEP 2020 Orientation & Sensitization Programme** by UGC-Malaviya Mission Teacher Training Centre Banaras Hindu University -2024
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Memberships

- Life Member – **Physics Society**, Central University of Allahabad
 - Life Member – **Indian Association of Physics Teachers (IAPT)**, Kanpur
 - Life Member – **Indian Meteorological Society**
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